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A Reconnaissance Study by Paleomagnetic Methods of the Age of Mineralization along the Viburnum Trend, Southeast Missouri

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Abstract

The Mississippi Valley-type lead-zinc ores of the Viburnum Trend in southeast Missouri are hosted by the Upper Cambrian Bonneterre Dolostone. The ore deposits are clearly epigenetic but conventional geologic and geochemical methods have failed to give an absolute date for the time of mineralization and thus have hampered the understanding of the ore genesis. The magnetic remanences of both ore and host are very weak but measurable. The paleopoles determined from the ore samples indicate an Early Permian-Late Pennsylvanian magnetic event. The host rocks collected in the mines have been remagnetized, showing paleopole positions similar to those of the ore samples. Correlative Bonneterre Dolostone samples collected far away from mineralization have poles quite different from those of similar host rocks in the mine; they probably still retain part of the original direction acquired during their sedimentation. To explain all of the paleomagnetic results in this study best, a mechanism for resetting the paleomagnetic clock by the mineralization is proposed. According to this mechanism, the age of the magnetic event observed from the rocks in the mine would probably represent the time of the mineralization. If this premise is accepted, the mineralization in the Viburnum Trend occurred in Early Permian-Late Pennsylvanian time.

The paleomagnetic methods used are regarded as being potentially applicable to many other problems of a stratigraphic and diagenetic nature. With the arrival of cryogenic magnetometers for routine usage, paleomagnetic studies of many additional deposit types will have come of age.

Introduction

THE stratigraphy and ores of the Viburnum trend of southeast Missouri are now well known from extensive drilling and mining, and the adjacent basins and Precambrian basement are also as well known as almost any others within the North American craton (*Economic Geology*, 1977, v. 72, no. 3). Despite this accumulated knowledge, neither geologic nor geochemical evidence has served to tie down the probable time of mineralization even within broad limits (just as the time of migration of oil and gas into traps commonly eludes the petroleum geologist). An earlier attempt to resolve this problem by paleomagnetic methods was made using a spinner magnetometer, but it was only partially successful because of the extremely low intensities to be measured and the small size of specimens that the machine would accept (Beales et al., 1974). However, with the development of the ultra high sensitivity cryogenic magnetometer in the early seventies, such dating problems can now be studied better, though still not without some inherent difficulties.

Although none of the major minerals of Mississippi Valley-type ores and host rocks are magnetic, entrapped or coprecipitated magnetic "impurities" may be. These impurities may be magnetized through various geologic processes, such as the formation of the host rock, metamorphism, or, as this paper attempts to show, mineralization. Thus, when the paleopole positions of the host rock and ore samples are compared to the apparent polar wandering path for North America, it is possible to determine the time when the paleomagnetic clock was finally set.

This paper records the results of reconnaissance paleomagnetic studies of the host rocks and ore of three mines along the Viburnum Trend, i.e., the St. Joe Minerals Corporation No. 28 mine in the north, Magmont mine in the center, and the Ozark mine in the south. In addition, two host rocks progressively more removed from the mineralized zone were studied to assist us in the interpretation of the "resetting" phenomenon.

Geology of the Viburnum Trend[1]

Although the mines differ in detail, they are treated here as local variations occurring along a thirty-mile-long, north-south, almost linear trend (Fig. 1). No basement, superficial rock linear fault, or fracture

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[1] Based largely on the Viburnum trend issue of *Economic Geology*, no. 3, 1977.

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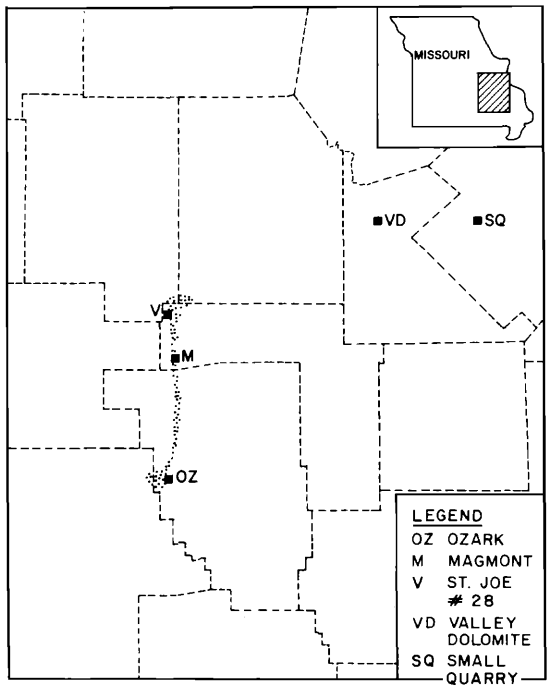


FIG. 1. Location map.

control has been mapped along the trend. Only a meandering north-south facies front persists between the environmentally more restricted and dolomitic veneer rocks of the St. Francois platform and the more open circulation limestones of the shallow basin to the west. It is approximately along the facies front that mineralization is concentrated. Major elongate, sublinear tracts of collapse breccia occur within the upper Bonneterre Dolostone, subparallel to the paleoshoreline. They have splayed boundary faults and are generally roofed by and incorporate sags of the overlying Davis Shale. These breccias were commonly the sites of higher than mine-average grade ore deposition, though in total tonnage, bedded disseminated ore in originally porous calcarenite facies is the main ore type.

In common with the Mississippi Valley group of ores as a whole, the ores occur in dolostones and, only to a minor extent, in equally porous and permeable sandstones. Also, fluid inclusion studies have shown repeatedly that the Mississippi Valley-type ores were probably deposited from low-temperature, high-salinity brines (Roedder, 1967, 1977a and b). The Viburnum mines, regionally and locally, are structurally high and on the flanks of a major sedimentary basin in a cratonic setting. The ore is stratabound, confined to the Upper Cambrian Bonneterre Dolostone, and of simple mineralogy, consisting predominantly of galena, sphalerite, marcasite, and pyrite. The mines are largely associated with drapes over highs of the hummocky Precambrian basement or within stratigraphic traps adjacent to monoclinal dips to the westerly trending basin. The lead isotopes are complex and have been regarded as derived from sedimentary sources (Doe and Delevaux, 1972, Sverjensky et al., 1979). Sulfur isotopes show a wide spread and are likewise considered to be derived from sedimentary sources, probably from sulfate deposited within more restricted facies of the Bonneterre host rock. There is no trace of metamorphism or any obvious igneous associations. Rarely, small pockets of trapped oil have been encountered during mining. The underlying porous and permeable Lamotte Sandstone overlaps the Precambrian basement which is an aquiclude. The feather edge of the Lamotte underlies the Bonneterre Dolostone which at the time of mineralization was certainly much more porous and permeable than it is at present and able to transmit the vast amount of metalliferous brine solutions required to form the orebodies. The overlying Davis Shale, also of Upper Cambrian age, acted as an aquiclude and restricted upward migration of fluids.

In contrast to many other Mississippi Valley-type ore deposits, influx of white sparry dolomite is minor, being more pronounced at the southern end of the trend. Other variations, such as unusually high copper values, also toward the south end, typify the local differences that are to be found along the trend. The complex histories of ore depositional events in differing areas clearly involved various tectono-environmental rock factors and probably the mixing of fluids from many sources. It becomes a matter of opinion which criteria to emphasize and which to regard as confusing overprints. The southeast Missouri area is here regarded as a type area of Mississippi Valley-type ore deposits and the precipitation mechanism presently preferred relates ore formation to the mixing of deep basinal brines carrying metals as soluble chloride complexes and formational fluids that provided a source of reduced sulfur. However, many problems remain unsolved, one of the most perplexing of which relates to the complex zoning of the otherwise simple mineralogy. This is well exemplified by the intramine and intermine differences in metals distribution along the Viburnum Trend. This zonation, particularly affecting Pb/Zn ratios and Cu and Fe distributions, is ubiquitous but poorly understood (*Economic Geology*, 1977). Variations in mineral overgrowth, leaching, brecciation, and rebrecciation suggest that ore-depositing fluids may have flowed along pathways that developed as evaporites and carbonates were preferentially dissolved from the sequence. Solution and local collapse areas would be influenced by the original deposition of more soluble minerals and structural fracturing. The mineralizing

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fluids probably also evolved and suffered changing pressure pulses through time. Another perplexing problem of ore emplacement relates to the timing of ore emplacement. The present paper reports efforts to use paleomagnetic imprinting to solve this aspect of the overall problem.

Instrumentation

The cryogenic magnetometer used in this study was constructed by Superconducting Technology Incorporated. The sensitivity of the two transverse sensors is 1×10^{-7} emu, the axial sensor has a sensitivity of 5×10^{-8} emu total magnetic moment; due to its double superconducting shield the signal is very stable. Our particular unit was a prototype and is probably the only one which is capable of accepting samples measuring up to 9 cm in diameter. Most samples were cut to about 7.5-cm cubes. Because of this unique feature, the lowest measurable magnetization is down to the 10^{-10} emu/g range. This sensitivity is about three to four orders of magnitude better than the conventional spinner magnetometer. Therefore, some weakly magnetized, commonly occurring rocks, such as limestones and dolostones, can be readily studied paleomagnetically. In order to determine the stable remanent magnetization direction either a thermal demagnetization or an alternating field demagnetization process was used. The whole demagnetization unit is housed in a μ-metal room in which the residual field can be kept in a 10 gamma range.

Paleomagnetic Measurements

All samples were cleaned by acid and then stored in the μ-metal room for several weeks. Demagnetization processes, mostly thermal demagnetizations, were carried out in the μ-metal room and each demagnetized sample was carried in a double μ-metal-shielded box to the cryogenic magnetometer. Here they were loaded into the sample holder assemblage above the magnetometer with a residual magnetic field of not more than a couple of hundred gammas; they were then inserted into the cryogenic magnetometer in less than 10 seconds. The reproducibility of the remanent magnetization measurements in emu/g on samples listed here is comparable to any reliable conventional paleomagnetic study.

Table 1 lists natural remanent magnetism and demagnetization data together with sample location and rock type. Admittedly, the total number of successful measurements is lower than in many other paleomagnetic analyses. This, in part, reflects the difficulty involved in the paleomagnetic analysis of Mississippi Valley-type host rocks and ores. The intensities of natural remanent magnetism for both ore and host are extremely low, usually down to the 10^{-8} emu/g level. In order to obtain a high magnetic signal to noise ratio we were forced to collect extra large samples. Unfortunately, pure ore samples of sizes larger than 10 cm^3 are difficult to find. Another difficulty involved is that many samples are viscous or become viscous during thermal demagnetization even at low temperatures.

Due to these difficulties, which were not fully appreciated when the sampling was done, we are unable to provide statistically as many data points as a conventional analysis normally does. However, it may be argued that the extra large sample size in some cases achieves a certain degree of sample averaging in itself.

To visualize the intensity and directional change during demagnetization of natural remanent magnetism, a normalized intensity versus a demagnetization temperature diagram and a stereoprojection of the declination and inclination change with demagnetization temperature can be constructed for each sample. As an example, Figure 2a and b show the intensity and directional changes, respectively, of the thermal demagnetization of the St. Joe No. 28 mine ore samples. Notice that both the intensity and direction of the natural remanent magnetism of samples VM7-4 and VM7-6 become steady and closely clustered up to about 350°C, then the intensity drops sharply to less than 20 percent of its original value. On the other hand, samples VM7-1 and VM7-2 show a gradual change of intensity first and then drop sharply above 300°C. Each natural remanent magnetism direction swings along a great circle path; they finally intersect with each other at the best estimate of a stable direction for samples VM7-4 and VM7-6. Sample VM7-8 has a lower signal to noise ratio, as shown in Table 1, than the rest of the VM samples. However, both its intensity and directional change are similar to samples VM7-1 and VM7-2. From these examples, a stable remanent magnetization component, shown by the intersecting directions in Figure 2b, clearly exists.

Technique for Paleomagnetic Remanence Direction Determination

It is not always easy to determine the remanent magnetization direction due to the multimagnetic components in these samples. To overcome this problem, a technique developed by Halls was used and a brief introduction of the method is described below (for details, reference can be made to Halls, 1976, 1978).

As a first approximation, it is assumed that each magnetic component in the sample carries a different direction. During thermal demagnetization the magnetization direction first moves away from the thermally unstable (low blocking temperature) phase along a great circle to the direction of the higher

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TABLE 1. Paleomagnetic Results*

Sample suite	Sample no. and wt. (g)	T.D. temp. (°C)	A.F.D. field (Oe)	Magnetic moment ($\times 10^{-5}$ emu)	Magnetization ($\times 10^{-8}$ emu/g)	Declination (degree)	Inclination (degree)	Signal to noise ratio
St. Joe No. 28 (ore), 91°7′ W, 37°43′ N	VM7-1	NRM		15.917	18.27	150.0	45.2	796
	871.3	160		14.054	16.13	150.1	38.2	703
		218		13.164	15.10	151.0	36.9	658
		269		10.687	12.27	154.6	30.5	534
		345		7.197	8.26	150.7	14.6	360
		392		2.754	3.16	147.5	7.0	138
	VM7-2	NRM		13.410	16.04	104.3	64.9	671
	836.1	160		11.841	14.16	114.2	61.1	592
		218		10.812	12.93	119.1	55.8	541
		269		9.082	10.86	130.4	44.8	454
		345		7.659	9.16	148.9	13.2	383
		392		1.891	2.26	147.3	13.8	95
	VM7-4	NRM		29.781	41.51	149.1	15.6	1,489
	717.4	160		30.959	43.15	150.2	9.8	1,548
		218		31.006	43.22	151.4	7.4	1,550
		269		30.632	42.70	149.8	5.3	1,532
		345		28.994	40.42	146.0	2.4	1,449
		392		6.929	9.66	189.8	−2.4	346
	VM7-6	NRM		37.843	42.678	138.0	34.3	1,892
	886.7	158		35.442	39.970	142.9	20.6	1,772
		204		36.394	41.045	142.2	17.6	1,819
		298		35.555	40.093	143.4	11.5	1,778
		364		31.556	35.558	146.7	5.2	1,577
		408		4.375	4.934	140.8	−13.7	219
	VM7-8	NRM		7.091	9.289	219.9	77.8	355
	763.4	158		5.125	6.714	204.8	76.0	256
		204		4.651	6.093	207.4	75.6	233
		298		1.126	1.475	143.1	−13.6	56
		364		1.963	2.570	118.6	−63.7	98
St. Joe No. 28 (host)	VH2-6	NRM		3.024	6.77	316.0	50.0	151
	446.6	150		2.613	5.85	330.0	12.0	131
		267		2.655	5.94	330.0	−4.0	133
		314		1.555	3.48	338.0	−9.0	78
		350		1.597	3.58	333.0	−12.0	80
		382		1.237	2.77	337.0	−9.0	62
		404		1.032	2.31	336.0	−9.0	52
		429		0.996	2.23	335.0	−11.0	50
		467		0.734	1.64	344.0	−13.0	37
	VH6-7	NRM		0.856	1.83	300.0	73.0	43
	468.8	150		0.604	1.29	329.0	28.0	30
		267		0.555	1.18	334.0	14.0	28
		314		0.318	0.68	337.0	−1.0	16
		350		0.246	0.53	328.0	−6.0	12
		382		0.208	0.45	331.0	−14.0	10
		404		0.118	0.22	322.0	−2.0	6
		429		0.188	0.40	347.0	−17.0	9
	VH2-1		NRM	2.772	6.275	317.0	50.0	139
	441.7		100	2.263	5.123	327.0	6.0	113
			200	1.879	4.256	315.0	23.0	94
			300	0.857	1.941	323.0	−2.0	94
			400	0.488	1.104	336.0	−8.0	24
Magmont (ore), 91°7′ W, 37°38′ N	MG-2	NRM		2.618	2.185	59.0	45.0	131
	1198.2	107		2.062	1.721	333.0	25.0	103
		181		1.964	1.640	333.0	15.0	98
		241		1.796	1.499	331.0	1.0	90

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TABLE 1—(*Continued*)

Sample suite	Sample no. and wt. (g)	T.D. temp. (°C)	A.F.D. field (Oe)	Magnetic moment ($\times 10^{-5}$ emu)	Magnetization ($\times 10^{-8}$ emu/g)	Declination (degree)	Inclination (degree)	Signal to noise ratio
		283		2.112	1.762	322.0	−1.0	106
		346		1.390	1.140	312.0	−2.0	69
		388		0.967	0.810	325.0	−18.0	48
	MG-3		NRM	2.871	1.635	314.0	39.0	144
	1756.0		25	2.290	1.304	318.0	34.0	114
			50	1.878	1.069	325.0	18.0	94
			75	1.960	1.116	331.0	8.0	98
			100	1.825	1.039	329	3.0	91
			125	1.768	1.007	338	0.0	88
			150	1.665	0.948	338	1.0	83
			200	1.220	0.695	339	−4.0	61
			250	0.789	0.449	330	7.0	39
Magmont (host)	ML-3	NRM		2.812	5.237	137	53.0	141
	537.1	107		2.148	3.999	145	13.0	107
		181		2.063	3.834	144	9.0	103
		241		1.938	3.609	147	6.0	97
		283		1.887	3.514	145	7.0	94
		346		1.816	3.381	145	11.0	91
		388		1.779	3.312	145	15.0	89
		490		1.927	3.588	134	16.0	96
Ozark (host), 91°10′ W, 37°24′ N	OZ2-1	NRM		2.568	5.880	111	76.0	128
	430.8	58		2.222	5.091	120	72.0	111
		90		1.700	3.893	132	62.0	85
		153		1.225	2.804	145	27.0	61
		214		1.082	2.478	161	−1.0	54
		266		1.355	3.102	155	−7.0	68
		319		0.913	2.091	156	−12.0	46
		350		0.911	2.087	156	−14.0	46
		382		0.275	1.341	143	−18.0	14
		401		0.842	1.930	172	−53.0	42
	OZ2-5	NRM		2.640	6.590	100	79.0	132
	400.6	58		1.453	3.626	341	58.0	73
		90		1.361	3.397	340	56.0	68
		153		1.249	3.118	338	52.0	63
		214		1.169	2.920	342	42.0	58
		266		0.958	2.392	337	39.0	48
		319		0.853	2.130	341	30.0	43
		356		0.581	1.451	332	21.0	29
		382		0.485	1.211	334	14.0	24
		401		0.407	1.016	364	27.0	20
		429		0.487	1.221	336	15.0	24
	OZ9-5	NRM		3.732	7.927	111	67.0	187
	470.8	58		3.404	7.231	121	60.0	170
		90		2.957	6.282	134	50.0	148
		153		2.694	5.722	148	24.0	135
		214		2.923	6.208	156	0.0	146
		266		2.957	6.280	158	−4.0	148
		319		2.397	5.090	159	−7.0	120
		350		2.005	4.258	157	−8.0	100
		382		1.290	2.740	154	−5.0	65
		401		0.837	1.779	153	−8.0	42
		429		0.451	0.958	149	7.0	23
	OZ9-6	NRM		3.876	7.734	103	68.0	194
	501.2	58		3.592	7.168	115	60.0	180
		90		3.067	6.120	124	50.0	153
		153		2.737	5.461	142	23.0	137
		214		2.926	5.837	150	1.0	146

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TABLE 1—(*Continued*)

Sample suite	Sample no. and wt. (g)	T.D. temp. (°C)	A.F.D. field (Oe)	Magnetic moment ($\times 10^{-5}$ emu)	Magnetization ($\times 10^{-8}$ emu/g)	Declination (degree)	Inclination (degree)	Signal to noise ratio
		266		2.995	5.976	152	−4.0	150
		319		2.376	4.740	153	−8.0	119
		350		2.157	4.304	156	−8.0	108
		382		1.306	2.605	159	−10.0	65
		401		0.928	1.851	155	−9.0	46
		429		0.443	0.884	148	8.0	22
	OZ9-7	NRM		3.098	7.40	289	69.0	155
	418.4	150		2.125	5.08	334	23.0	106
		267		2.320	5.54	339	−2.0	116
		314		1.140	2.73	340	0.0	57
		350		1.056	2.53	342	−7.0	53
		382		0.885	2.12	337	−14.0	44
		404		0.559	1.34	339	2.0	72
		429		0.446	1.07	342	21.0	22
		467		0.392	0.94	349	−21.0	20
Small quarry (host), 90°19′ W, 37°55′ N	SQ1-1	NRM		5.609	8.996	9.0	72.1	280
	623.5	156		1.561	2.504	96.7	60.2	78
		218		1.302	2.087	114.1	54.3	65
		268		1.143	1.834	116.8	50.5	57
		320		1.008	1.617	120.8	46.6	50
		372		0.662	1.062	122.9	30.4	33
		400		0.530	0.849	110.2	78.9	27
	SQ1-2	NRM		5.847	8.397	357.9	69.9	292
	696.3	158		1.517	2.178	94.6	67.1	76
		204		1.307	1.877	105.7	60.7	65
		298		0.925	1.328	123.0	37.9	46
		364		0.816	1.172	135.3	37.6	41
		408		0.994	1.428	153.5	36.6	50
	SQ2-1	NRM		5.467	9.519	39.1	65.5	273
	574.3	156		1.427	2.484	17.8	65.2	71
		218		0.892	1.553	27.0	69.4	45
		268		0.645	1.123	29.7	68.4	32
		320		0.243	0.423	126.3	56.3	12
		372		0.157	0.273	106.9	−67.5	8
	SQ2-2	NRM		4.647	7.932	14.7	66.3	232
	585.9	156		0.757	1.291	49.1	74.8	38
		218		0.475	0.810	107.1	73.4	24
		268		0.320	0.545	117.5	71.2	16
		320		0.164	0.280	130.9	58.5	8
		372		0.120	0.205	222.8	−40.1	6
	SQ2-3	NRM		5.382	9.130	12.2	65.8	269
	589.5	156		1.097	1.860	26.3	62.8	55
		218		0.558	0.947	48.7	65.3	28
		268		0.443	0.752	52.6	66.8	22
		320		0.168	0.285	63.9	65.0	8
		372		0.278	0.471	242.5	−43.8	14
	SQ3-1	NRM		3.563	5.664	194.7	60.7	178
	629.0	154		0.773	1.229	169.9	62.4	39
		269		0.183	0.291	119.1	45.3	9
		320		0.281	0.446	186.7	−52.4	14
		388		0.183	0.290	91.0	−41.1	9
	SQ3-2	NRM		3.168	5.137	198.0	57.2	158
	616.7	154		0.926	1.501	189.2	58.3	46
		269		0.204	0.331	135.7	45.4	10
		320		0.110	0.179	205.2	−64.8	6
		388		0.145	0.235	45.0	71.5	7

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TABLE 1—(*Continued*)

Sample suite	Sample no. and wt. (g)	T.D. temp. (°C)	A.F.D. field (Oe)	Magnetic moment ($\times 10^{-5}$ emu)	Magnetization ($\times 10^{-8}$ emu/g)	Declination (degree)	Inclination (degree)	Signal to noise ratio
Valley Dolomite (host), 90°32′ W, 37°55′ N	VD2-1	NRM		2.300	7.143	118.9	7.9	115
	322.0	154		2.121	6.587	120.3	2.4	106
		266		1.458	4.528	137.7	−7.3	73
		319		1.096	3.404	139.6	8.5	55
	VD2-2	NRM		2.592	6.894	142.1	15.7	130
	376.2	154		1.945	5.173	142.5	16.1	97
		266		2.139	5.689	143.8	−7.9	107
		319		1.706	4.537	142.2	−12.2	85
		382		1.120	2.979	178.2	−45.8	56
		450		0.546	1.452	7.8	−59.6	27
	VD2-6	NRM		1.073	3.040	132.8	34.0	54
	353.1	154		1.167	3.306	134.4	19.3	58
		266		0.947	2.683	141.0	5.9	47
		319		0.638	1.807	120.7	−3.6	32
	VD2-8	NRM		1.223	3.351	138.6	32.1	61
	364.8	154		1.014	2.778	136.8	23.2	51
		266		0.974	2.668	135.5	18.6	49
		319		0.909	2.490	137.2	25.4	45

* The Ozark and Magmont ore samples were essentially massive galena, the St. Joe No. 28 ore samples reported were highly marcasitic with minor galena. All the host-rock samples were medium- to fine-grained sucrosic dolomite, some showing traces of original calcarenitic and/or stromatolitic texture on polished slabs, except for the Small quarry samples which were fine-grained dolomitic to highly dolomitic limestones. In addition to the results presented about 20 samples had to be discarded either because they were too viscous (12), the instrument was discovered to be out of adjustment (3), or critical records had been obliterated (5). About another 30 samples had very little or no natural remanent magnetization and were not considered worth demagnetizing. T.D. = thermal demagnetization; A.F.D. = alternating field demagnetization; NRM = natural remanence magnetization

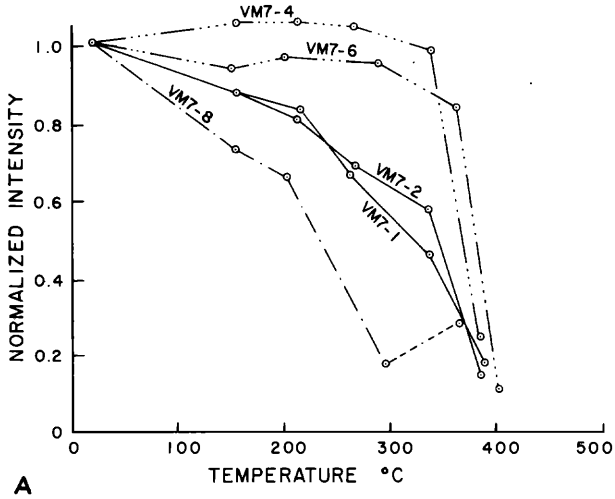
blocking temperature phase. Upon further thermal demagnetization, the magnetization direction will move along a different great circle to the direction of the highest blocking temperature phase. Utilizing a Lagrangian multiplier method (input, the demagnetization data; e.g., Table 1), a computer program can find the pole to the plane defined by the best-fitted least-squares great circle from the demagnetization data. A quality factor Q is calculated to evaluate the degree of goodness of fit. Here $Q = \lambda^2_{min}/(\lambda_{int} \cdot \lambda_{max})$, in which the λ's are the three eigenvalues obtained from the least-square fit. Q is a dimensionless parameter that can range from 0 to 1, corresponding to perfect fit or random direction, respectively. Halls suggested that an empirical reference value of 0.2×10^{-3} can be chosen. If Q exceeds this value, demagnetization data normally will be rejected. If, for instance, each sample from the same site has the same stable direction (possibly carried by magnetite) but different soft component directions (possibly due to later alteration), all the best-fitted great circles from each of these samples should intersect or converge at the stable direction.

To find this end point direction, first use the above-mentioned program to find, for each sample, the pole to the great circle. Now, if all the great circles do converge, the poles themselves should lie on a great circle and the pole to this last great circle is identical to the converging end point. Then, use the program again and put in the pole positions of samples from the same site as the points on a great circle to find the statistically best-fitted pole to this great circle. This last best-fitted pole is then interpreted as the end point of the stable phase. The Viburnum material is far from ideal, but a Q factor can also be calculated as a measure of the reliability of the end point determination. Normally $Q = 0.5 \times 10^{-3}$ is used for reference. If Q is lower than this value, the end point determined is considered reliable.

Again taking the VM samples from the St. Joe No. 28 mine as examples, each of the five ore samples move away from unstable directions along a great circle path as shown in Figure 3. The pole to each of the proposed five great circles and the best-fitted great circle to these five poles are also shown in Figure 3. Applying the same computer program, the pole to this last great circle is calculated to be at declination 147.1° and inclination 5.5° (shown in Fig. 3). The low Q factor of the 5.5×10^{-5} calculated suggests that this is a reliable estimate of an end point direction.

The purpose of this paleomagnetic study is to determine the paleopole positions of the host and ore

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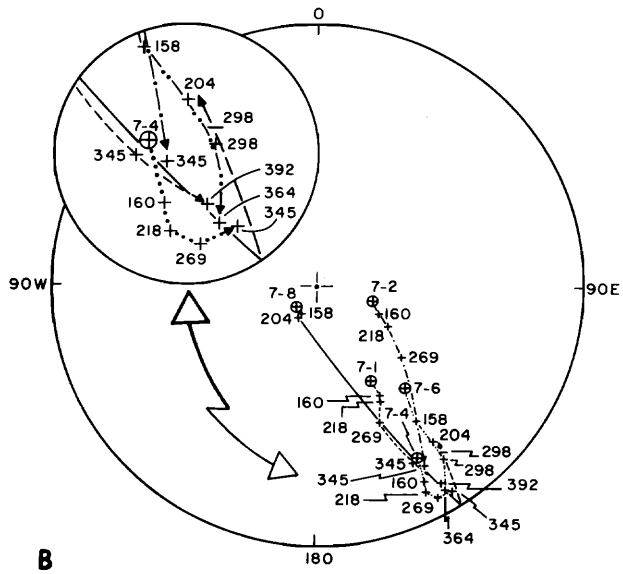


FIG. 2. A. Thermal demagnetization of VM7 samples 1, 2, 4, 6, and 8 (see Table 1 for sample notations). B. Directional change of VM7 samples 1, 2, 4, 6, and 8 during thermal demagnetization. Northern hemisphere is denoted by +. All directions at various temperatures are transformed to equivalent paleopole positions and plotted here for the convenience of comparison.

samples. Although the end point directions determined are close to the estimates by conventional methods, we think that Halls' method is statistically sound and useful for this multicomponent case. Table 2 lists all of the calculated directions of the best-fitted pole to the great circle of each sample and the direction of the end point of a suite of samples in terms of declination and inclination estimated by the Halls' technique. The only exception is the Magmont host where ML3 is the only sample available. Fortunately, this particular sample is extremely stable both in direction and intensity; the estimate of the stable end point direction is found by averaging the demagnetization data from 107° to 388°C, above which measurements became erratic.

The paleopole position of each suite of samples can then be calculated from the location of the sample site and the end point direction of that suite of samples. Table 2 lists the paleopole positions of the ore samples from the St. Joe No. 28 mine and the Magmont mine, as well as the host samples from the St. Joe No. 28, Magmont, and Ozark mines. In addition, the paleopoles determined from host samples well away from mineralization at the Small quarry and the Valley Dolomite quarry are also included. The next section compares these paleopole positions to others from North America to estimate their apparent ages.

Apparent Ages Determined

Recently, Irving (1979) reviewed the published paleopole positions for the stable North American craton. His lists of the average paleopole positions that he used to calibrate an apparent polar wandering path clearly indicate that the portion of the path from late Carboniferous to the present is well established. This is demonstrated by the number of values (N), the precision parameters (K), and the radii of the circles

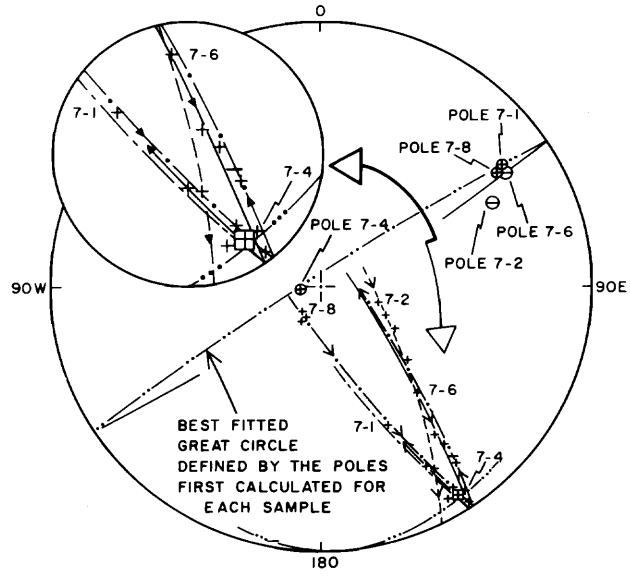


FIG. 3. An example of an application of the great circle method of Halls (1976, 1978) applied to VM7 samples using the higher blocking temperature range, i.e., magnetite. The dashed curves represent best-fitted great circles calculated by the Lagrangian multiplier method for each individual sample. The solid curve represents the best-fitted great circle to the calculated poles of each sample from the same site. The pole to this solid great circle is denoted as the square with the cross which, therefore, represents the direction of presumed magnetite phases in Viburnum ore.

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TABLE 2. Stable End Point and Paleopole Position Determinations, Viburnum Trend, Southeast Missouri

Sample suite	Sample no.	Best fitted pole				Stable endpoint direction			Paleopole position determined	Long.	Lat.	Polarity
		Decl.	Incl.	Q × 10^E	Polarity	Decl.	Incl.	Q				
St. Joe No. 28 (ore), 91°7′ W, 37°43′ N	VM7-1	56.1	14.4	0.1E-6	R							
	VM7-2	63.2	−20.0	0.2E-6	R							
	VM7-4	−95.5	81.5	0.3E-5	R	−32.9	−5.5	5.5×10^{-6}	VM	127.9 E	42.4 N	R
	VM7-6	57.2	−13.3	0.7E-6	R							
	VM7-8	55.7	12.4	0.2E-7	R							
St. Joe No. 28 (host)	VH2-6	68.6	13.8	0.2E-4	N							
	VH6-7	72.2	10.7	0.2E-6	N	−22.5	−8.1	5.9×10^{-4}	VH	114.8 E	45.1 N	N
	VH2-1	60.2	−21.0	0.3E-7	N							
Magmont (ore), 91°7′ W, 37°38′ N	MG-2	61.2	−4.7	0.8E-7	N							
	MG-3	66.0	61.3	0.1E-6	N	−27.7	1.7	1.1×10^{-7}	MG	124.7 E	47.4 N	N
Magmont (host)	ML-3					145.2	10.2		ML	129.2 E	38.8 N	R
Ozark (host), 91°10′ W, 37°24′ N	OZ2-1	74.3	43.2	0.2E-3	R							
	OZ2-5	61.4	−10.0	0.5E-4	N							
	OZ9-5	−159.0	80.2	0.1E-3	R	−25.8	7.2	1.8×10^{-6}	OZ	124.0 E	50.8 N	M
	OZ9-6	−129.0	66.9	0.7E-5	R							
	OZ9-7	63.2	−0.5	0.2E-4	N							
Small quarry (host), 90°19′ W, 37°55′ N	SQ1-1	60.1	−24.0	0.1E-5	R							
	SQ1-2	44.1	−14.0	0.2E-7	R							
	SQ2-1	66.9	−16.0	0.4E-5	R							
	SQ2-2	59.3	−10.0	0.2E-6	R	−51.6	−43.4	4.6×10^{-4}	SQ	132.7 E	12.7 N	R
	SQ2-3	54.5	−24.0	0.1E-5	R							
	SQ3-1	−3.3	27.8	0.5E-8	R							
	SQ3-2	6.3	31.9	0.4E-8	R							
Valley Dolomite (host), 90°32′ W, 37°55′ N	VD2-1	−143.0	53.3	0.16-4	R							
	VD2-2	41.7	10.9	0.3E-5	R	−44.3	1.8	1.7×10^{-4}	VD	144.3 E	37.8 N	R
	VD2-6	46.3	−33.0	0.1E-3	R							
	VD2-8	50.7	−15.0	0.1E-7	R							

of confidence (α_{95}) given for each pole position. Unfortunately the path prior to 300 m.y. ago is not as well defined as the more recent portion. Therefore, apparent age determinations based upon the apparent polar wandering path can only be regarded as best guesses compared with the greater reliability assigned to rocks younger than about 300 m.y. ago.

From Table 2, the paleopoles VM, VH, MG, and OZ are very close to each other. If we accept the resetting mechanism proposed in the next section, their mean position would indicate a paleomagnetic event. The mean position (MV) is found at latitude 45° N and longitude 124° E. The Fisher statistics calculation indicates a high precision parameter K of 176 and an acceptable α_{95} of 5.8°.

Figure 4 compares the paleopoles determined in this study to the apparent polar wandering path of North America craton. An Early Permian-Late Pennsylvanian apparent age can be estimated for the pole MV. On the other hand, paleopole SQ is much farther south toward a possible Late Silurian pole position at 400 m.y. The paleopole VD is somewhat east of the Pennsylvanian pole position, but neither of these determinations is securely dated.

Two most important conclusions can be drawn from the paleopole position data. First, the paleopole positions of the host samples collected in the mines (i.e., VH, OZ, and ML) are very close to those of the ore samples (i.e., VM and MG). Their mean position MV has an apparent age of Early Permian-Late Pennsylvanian time. Second, the paleopole position of the host sample far away from mineralization (i.e., SQ) is very different from those of samples collected in the mines (i.e., VH, OZ, and ML).

Resetting of the Paleomagnetic Clock by Mineralization

The conclusions stated in the previous section raise two interesting questions. First, why, on the one hand, do the host samples in the mine have paleopole positions similar to those of the ore samples, whereas, on the other hand, host samples far from mineralization have paleopole positions quite different from those in the mine? Second, what is the relationship

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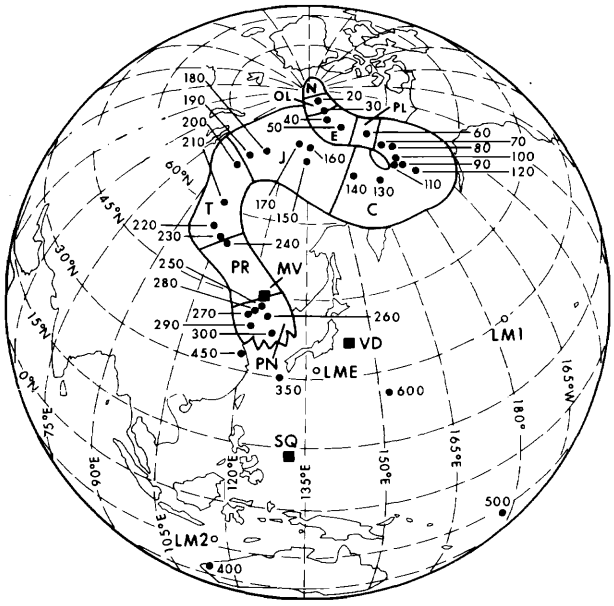


FIG. 4. Comparison of paleopoles determined in this study to the apparent polar wandering path of North America (slightly modified, after Irving, 1979); the number by each paleopole (solid circles) is the estimated age in millions of years. The envelope of the path suggests the α_{95} of these points. Due to the insufficient number of published paleopoles older than 300 m.y., no envelope is shown in this diagram for that portion of the path. The symbols in the envelope of the apparent polar wandering path are: N = Neogene, OL = Oligocene, E = Eocene; PL = Paleocene, C = Cretaceous, J = Jurassic, T = Triassic, PR = Permian, and PN = Pennsylvanian. The solid squares are paleopoles determined in this study. Pole MV is the average pole position of the five sample sites in the mines (VM, VH, MG, ML, and OZ). SQ is the pole of the host samples from the small quarry far away from mineralization. VD is the pole of the Valley Dolomite host samples within the Old Lead belt but not immediately adjacent to ore.

The open circles are Lamotte Sandstone paleopoles determined by Al-Khafaji and Vincenz (1971): LM1 is their group 1, LM2 is group IIb, LME is group E; their group 11a, 38° S 168° W, not shown on this projection, compares closely with the St. Joe No. 8 mine pole position, 38° S 170° W, obtained by Beales et al. (1974).

between the magnetic event and the mineralization? In this section we shall comment on these questions and then propose a resetting mechanism.

There are several processes whereby these rocks may have been magnetized. The limestone precursor of the dolostone host rock may have been magnetized initially by acquiring a detrital remanent magnetization during sedimentation under the influence of the ambient geomagnetic field (Beales et al., 1980). That is, the fine-grained magnetic particles in a sedimentary rock source may be statistically aligned along the direction of the ambient field under suitable depositional conditions. As the sediment is consolidated, this direction is locked in place, or fossilized in the rock. Thus, the paleomagnetic clock is set initially at the formation of the host rock. However, the original detrital remanent magnetization may be partially or even completely remagnetized by any of the following processes: (1) by metamorphism, which may produce so-called thermoremanent magnetization or Piezo-remanent magnetization; (2) by events such as lightning, shock, or simply the viscous effect of the magnetic particles in a changed ambient field; and (3) by chemical processes such as weathering, cementation, and redox reaction, which may alter the original remanent magnetization and produce chemical remanent magnetization. One variety of this might be the type of mineralization which we propose in the following section.

As stated above, typical Mississippi Valley-type lead zinc ores have no trace of metamorphism or any obvious igneous association. Fluid inclusion studies have shown repeatedly that this type of ore was probably deposited in a low-temperature environment (mainly in the range 94° to 120°C, in Viburnum Trend, according to Roedder, 1977a). Therefore, remagnetization due to metamorphism is not likely. Other remagnetization events are doubtful, too. Lightning may certainly cause partial remagnetization; however, it is soft and can be easily eliminated in the process of demagnetization. Furthermore, its direction usually reflects the present geomagnetic field direction. Shock events also are almost impossible, since there is no evidence of them in the ore deposits. Although in some unusual conditions viscous remanent magnetization may be stable against demagnetization, it would be unlikely to produce a consistent stable direction from mine to mine.

Chemical processes such as weathering usually do not affect the rocks found in the mine. That they may be important on the surface has been observed in VD samples. The quarry from which these samples were collected is located closer to the crest of Saint Francois Mountains and certainly has been subjected to groundwater penetration. The pervasive pale rusty brown coloration of these rocks and solution channels suggests surface karst involvement of these samples at some time. The VD paleopole position (see Fig. 4) is unmistakably different from both the MV and SQ poles. This suggests that the development of a chemical remanent magnetization could have occurred, probably several times, since the formation of the Bonneterre Dolostone. However, the scale of the resetting events needed to account for the consistent paleopole positions we observed from the host rocks and ores in the mines probably excludes localized solution and cementation of this kind.

Finally, since the host dolostone samples from the mine have paleopole positions indicating the same Early Permian-Late Pennsylvanian age, and since the Bonneterre Dolostone was formed in Late Cambrian times, this means that the original detrital remanent

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magnetization acquired during sedimentation was probably negligible or unlocked during diagenesis and the rocks in the mine were then remagnetized (Fig. 5).

The above considerations are by no means exhaustive but simply mean that we need a mechanism to explain them.

The simplest model is clearly that the magnetization or a remagnetization process was directly related to mineralization.

In this context, the mechanism of the paleomagnetic resetting process in sediments is certainly important to these studies. Verosub (1977), in his review of 40 years of research on the processes by which a sediment acquires a detrital remanent magnetization, emphasized the importance of postdepositional processes which arise from the mobility of magnetic carriers within fluid-filled voids in the sediments. Experimental results demonstrated that magnetic remanence will be set at the time when magnetic particles entrapped in fluid-filled voids lost their mobility, either by dehydration (Løvlie, 1974; Irving and Major, 1964; Irving, 1957; Khramov, 1968) or through gravitational compaction or cementation (K. Yaskawa, written commun., 1974). The mechanism of this paleomagnetic lock-in process is probably complicated, as stated by Meade (1966). Microscopic properties, such as size, shape, and composition of both the sediment and of the magnetic particles, will all determine the locking process.

Regardless of the complications of this locking process, it is clear that during mineralization, the Bonneterre Dolostone in the mineralized area must have been very much more porous and a vast volume of fluid must have passed through it to transport the tens of millions of tons of lead present in the galena deposits of southeast Missouri. With such a highly porous plumbing system, some magnetic particles, such as a minor impurity either in the host rock or carried in the brine, will certainly be free to rotate in the larger voids or cavities wherever the brine migration path cut through during mineralization. When the magnetic particles were trapped in the ore crystal or when postdepositional diagenetic processes last affected the ore and adjacent host rocks, the aligned magnetic particles were then locked in position. Therefore, they acquired a postdepositional remanence component as an overprint in addition to any of the original detrital remanence which was still carried by the magnetic impurities in the untouched portion of the host rock. Since the postdepositional remanence component may well be the predominant one of the natural remanent magnetism, then the paleomagnetic clock was probably reset and the simplest hypothesis is that this occurred during the mineralization.

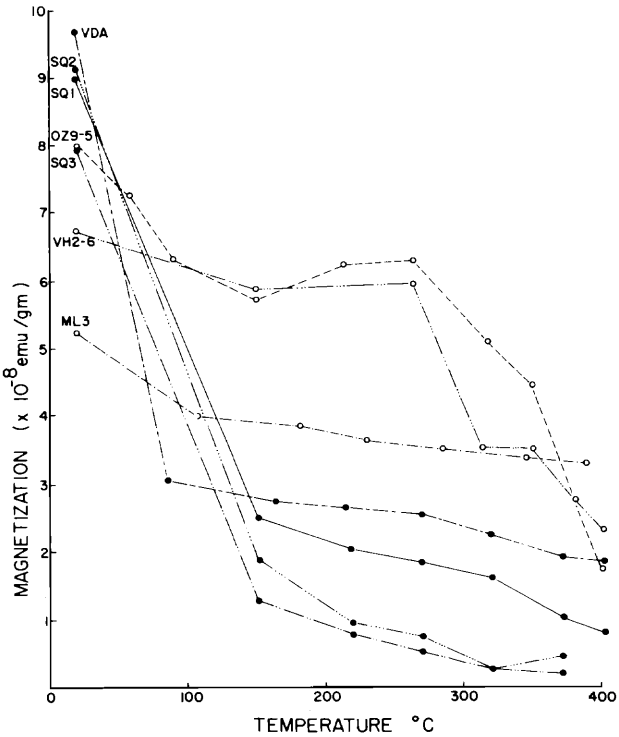


FIG. 5. Comparison of magnetization between host rock near a mineralization zone (open circles) and host rock far away from the zone (solid circles).

Figure 6 illustrates the proposed mechanism. During the formation of Bonneterre Limestone in Late Cambrian times (Fig. 6A), the fine detrital magnetic particles became aligned to the ambient magnetic field, direction H (notice the low inclination). Possibly during dolomitization and/or further cementation and dehydration during compaction, these particles became locked into place and subsequent changes of the geomagnetic field will not affect them unless they should become unlocked again by solution. Later, during the time of mineralization (Fig. 6B), the geomagnetic field changed (note the higher inclination). The large flux of brine solution, carrying or forming new minute magnetic particles, passed through the plumbing system, i.e., the Bonneterre Dolostone. In the brecciated region where the ore precipitated in large open cavities (area 3, enlarged in Fig. 6C), magnetic particles were precipitated or entrapped and enclosed (locked) by successive mineral growth; these would carry the same chemical, or detrital, remanent magnetization as the direction of the new field. However, in the part of the host rock through which the brine solution passed (area 1 and 2, enlarged in Fig. 6D), released magnetic particles carried in the brine solution were free to align with the new field, giving a reset detrital remanent magnetization.

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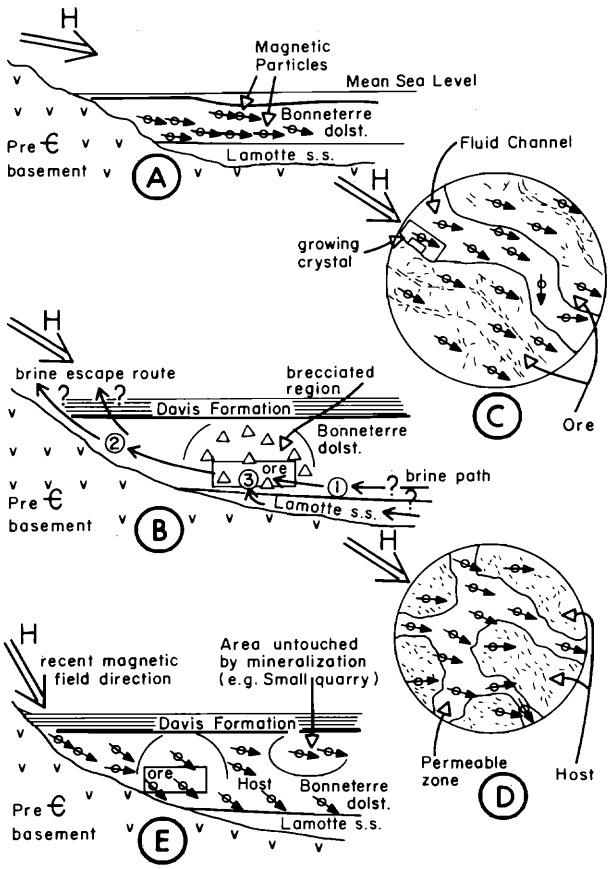


FIG. 6. Model (not to scale) for the overprinting process resetting the paleomagnetic clock in the Virburnum Trend area. A. The acquisition of detrital remanent magnetization in the host rock during its formation. B. The remagnetization process during mineralization. C. An enlarged view of area 3 in (B), showing the acquisition of detrital remanent magnetization in the ore during the mineralization. D. An enlarged view of areas 1 and 2 in (B), demonstrating the realignment of magnetic particles in the host rock near an ore deposit. E. The fossilized direction in the host rock after mineralization ceased. This locked direction is untouched by the more recent magnetic field change.

Toward the end stage of the mineralization (Fig. 6E), when voids were filled with new precipitation and fluid migration ceased, the new field direction was locked again in the host rock. This new field direction would be untouched by later changes of the geomagnetic field. The magnitude (intensity) of this component dominates the earlier rock magnetism, and the reset resultant that we measured is thus approximately parallel to the new field direction.

According to this mechanism, the age of the mineralization in the Viburnum Trend would be around Early Permian to Late Pennsylvanian, i.e., just within the more reliable portion of the apparent polar wandering path.

Discussion

The results here are interesting for a variety of reasons and only more work will answer some of the problems raised. However, some geological considerations will be discussed in this section, first, about the proposed mechanism, then, about the reinterpretation of earlier work, and, finally, about the ore precipitation.

The proposed mechanism assumes that the paleomagnetic clock has not been reset since mineralization. For these Mississippi Valley-type deposits, this assumption appears to be reasonable because the host rock has never been metamorphosed, and it occurs in the tectonically relatively inactive south central North American craton. The mechanism supposes the corollary that the ore and nearby host will generally have similar paleopole positions, whereas the host rock well removed from the mines may show an original detrital remanent magnetization or some secondary directions, e.g., thermoremanent or chemical remanent magnetization unrelated to the mineralization. As noted in Figure 4, paleopole SQ is more than 30° south of the average paleopole MV of host rock in the mines. Although the apparent polar wandering path earlier than Carboniferous times is not defined due to the inconsistent and insufficient published data, SQ is most likely much older than MV and probably reflects a lower Paleozoic pole.

Other host samples (VD) from the Valley Dolomite quarry, as mentioned earlier, seem to present a more complicated case. This type of complication may prove to be something to expect in the vicinity of orebodies where fluid migration routes were diverse and where some parts of the host rock may retain the original detrital or early diagenetic remanence whereas other parts may have been reset by diagenetic and epigenetic processes later than the mineralization itself.

There is a vague indication that the average intensity of the stable component of the host rocks in the mines is somewhat higher than that of samples collected far away from the mineralization. Figure 5 shows such a comparison. The OZ, ML, and VH samples, for instance, have more than twice the intensity of the SQ and VD samples after demagnetization at 250°C. If this is true, this may impose a constraint to the proposed model. That is, for the resetting mechanism to be effective, the magnetic particles in the influx brine solution should either be more stable or have a higher concentration than those originally in the host rock.

In light of the proposed mechanism, some results published earlier may now require review. Beales et al. (1974) suggested that the Bonneterre Dolostone and the ore at the St. Joe No. 8 mine of the Old Lead

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belt were Late Cambrian in age. This interpretation depended heavily on some of the Lamotte Sandstone results reported by Al-Khafaji and Vincenz (1971). The results from the St. Joe No. 8 mine (170° W 35° S, geomagnetic pole) agree with some of those from the Lamotte Standstone exposed around the Saint Francois Mountains (168° W, 38° S, for the group IIa of Al-Khafaji and Vincenz, 1971). Figure 4 shows these earlier results together with the results from this study. It is possible that the Lamotte Formation, hitherto regarded as largely representing Late Cambrian paleomagnetic events, is more extensively remagnetized than Al-Khafaji and Vincenz suspected at the time their work was written. Resetting of Lamotte paleomagnetic imprints is consistent with the Lamotte, an excellent aquifer to this day, being a major channel for basinal fluid transmission and, therefore, a potential site for emplacement or chemical alteration of magnetic impurities at the same time. This could explain some of the wide scatter in results re-

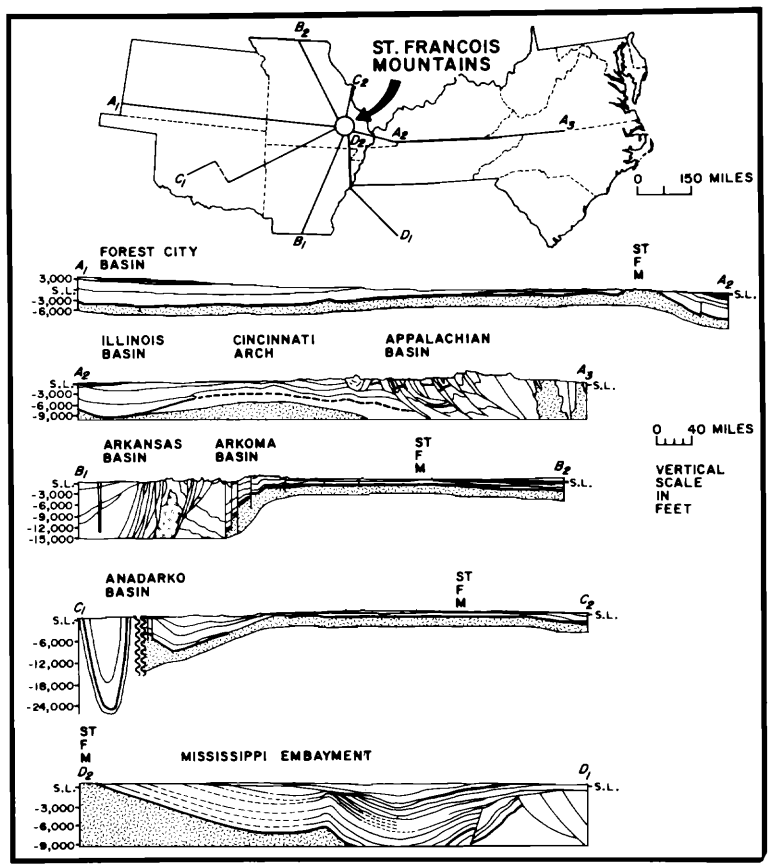


FIG. 7. Stratigraphic position of the Bonneterre Dolostone and equivalent strata (modified after the American Association of Petroleum Geologists, Geological Highway map series). Relative sediment loading that might develop a fluid drive system due to compactional or thermal difference, or comparison of tectonic settings provides a crude means of assessing likely fluid migration paths assuming that at least the sandstones (Lamotte Sandstone in southeast Missouri) underlying the Bonneterre host rocks maintained good aquifer permeability. (The Lamotte-Bonneterre unit is emphasized by the heavy line on the cross sections.)

A1–A3 = westerly transport route across the Forest City basin and easterly transport route from the Appalachian front across the Cincinnati arch and Illinois basin to flanks of the Ozark uplift—Old Lead belt. B1–B2 = southerly route from the Ouachita front across the Arkoma Basin to southern flank of the Ozark uplift and northerly route across the northern Forest City Basin. C1–C2 = southerly route from Anadarko basin across the Nemaha uplift-Seminole arch and central platform to the Ozark uplift, northerly route from nothern extension of the Illinois basin. D1–D2 = southerly route from the Appalachian front (buried beneath the Mississippi embayment) across central platform to southern flank of the Ozark uplift. N. B. The diagrams have only stratigraphic validity; actual fluid migration routes would be vastly more complicated depending on subsurface pressure gradients and major permeability trends, commonly fracture controlled.

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ported by Al-Khafaji and Vincenz, who also noted that a difficulty they encountered was that the Lamotte intensities measured varied significantly, which is a characteristic to be expected where remanence is largely due to introduced magnetic impurities or where resetting is achieved by overprinting by a larger amount of subsequently introduced material. Finally, it is possible that the main mineralization of at least part of the Old Lead belt is older than that along the Viburnum Trend, but, in view of proven resetting in the Bonneterre Formation, Late Cambrian is now viewed with some uncertainty until more work has been done.

If the mineralization did occur in Early Permian-Late Pennsylvanian times, we can consider the evolution of the areas around the Ozark uplift during the Paleozoic especially with regard to the possible time or times for the migration of brine solutions. If precipitation of the ore deposits depended on admixture of the metal-carrying brines with a fluid source of reduced sulfur, it can probably be linked to the maturation and migration of a considerable amount of oil or gas. This would be best facilitated by either burial of the source areas to depths of many thousands of feet or by their exposure to a local high heat flow. Possible source areas and likely times of subsurface fluid migration for the Viburnum area can therefore be assessed. Schematic cross sections (Fig. 7) are given for the Forest City basin to the west, the Arkansas-Anadarko basins to the south and southwest, and the Illinois basin to the east. Assuming that the most common fluid migration routes in sediments in cratonal settings are mainly lateral, consideration of these cross sections may indicate why a considerable time elapsed between host-rock formation and ore emplacement. All of the basins lacked thick sediment cover until the Pennsylvanian increment was added, and, despite its proximity to the Viburnum Trend, the Forest City basin appears least likely to have developed a brine and petroleum fluid maturation-migration system of its own. This does not preclude the possibility that porous and permeable strata in this basin were pressurized by sedimentary and tectonic events to the south. Coincidence of very thick sediment loading and orogenic events in the Ouachita geocline (Sutherland and Manger, 1979) in late Carboniferous times certainly made this area the most likely fluid source despite its considerable distance from Viburnum. The Saint Francois highlands were the high point of the Ozark dome and were probably undergoing erosion and warping during Late Pennsylvanian times. Also the area may have been affected by minor reactivation of basement faults and fractures that was reflected in fracturing and flexuring of the cover rocks. These factors rendered this area a likely break out point for fluids from a pressurized aquifer system—quite possibly the Lamotte Sandstone. This intrastratal flow was probably the cause of solution collapse and porosity enhancement in the Virburnum Trend area. Possibly local solution of evaporites and oil entrapment accompanied this, though a still earlier surface karst system may have enhanced porosity-permeability development in the area and thereby contributed to the likelihood of fluid breakthrough in the Saint Francois area. If anything, our paleomagnetic studies favor a purely intrastratal solution for porosity enhancement at Viburnum because we see no indication of a pre-Late Pennsylvanian karstification event leaving an overprint on mine host-rock paleomagnetics as may have happened with the Valley Dolomite. Leach (1973), in a study of barite-lead-zinc deposits of central Missouri, concluded that these deposits were probably of Late Mississippian to Pennsylvanian age and that they were also derived from the Ouachita geocline. Noting the absence of barite in the southeast Missouri lead district, he considered it to be genetically dissimilar. The present study raises the possibility that the two areas are related sequentially; earlier or later fluid escape routes may have carried metalliferous brines with abundant barium which was lacking at the stage of basin evolution during which the Viburnum Trend was acting as a major escape route. Alternatively, the timing may have been the same, but the presence of barite in the one case and not in the other may reflect differences in the escape routes traversed or depletion to absence of the sulfate precipitant.

Considering the timing of the mineralization and the trend of the orebodies, one may hypothesize that north-south stress developed during plate interactions associated with the Ouachita fold belt. These caused regional but minor, mostly lateral, movements in the basement throughout the evolution of the Viburnum area. Such movements could be reflected in the linearity of the sedimentary and evolutionary features of the Viburnum Trend area. Thus basement structural influences, combined with the development of a gentle hinge line along this trend, probably controlled the sublinear north-south facies front at the host-rock depositional stage. Associated barrier bar-restricted salinas could have developed behind the frontal higher energy sedimentary facies and emplaced highly soluble evaporites. Later, possibly in Pennsylvanian times, basement warping, combined with the onset of orogeny, in the Viburnum area caused tensional fractures in the cover rocks that opened up permeability routes. Preferential solution of the more susceptible rocks along the same trends took place, developing the brecciated host and enhancing permeability, while at the same time re-

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moving all traces of the former evaporites (Beales and Hardy, 1977). By latest Pennsylvanian times conditions were ripe for precipitation of the ores.

Ore Genesis

With the proviso that we regard a subsurface mixing of basinal brines, carrying the metals, with reduced sulfur from host-rock formational fluids as the most likely sulfide precipitation mechanism, (Anderson, 1975, 1978; Beales, 1975),[2] a possible synthesis of the paleomagnetic interpretations and Viburnum ore field geology may be stated briefly as follows:

Somewhat before the onset of precipitation, the Saint Francois Mountains would have been an area of minimal vertical confining pressures due to overlying strata at the time of this pressurization. Therefore, it would be a likely place of breakthrough and escape of large volumes of pressurized subsurface fluids. Possibly intrastratal solution, at least in part of evaporites, created reservoir conditions for oil and gas at points of structural and stratigraphic closure. Sulfate reduction by organic reductants generated H_2S and possibly S which in turn acted as the precipitant of the ore minerals. That these interrelated events logically occupied a considerable period of later Paleozoic time is supported by the mixed, antipodal nature of the measured paleomagnetic directions from samples cut from the same block of ore. This suggests that, whatever the accuracy of the assigned paleomagnetic timing, mineralization lasted at least for the duration of a single paleomagnetic reversal; this may mean a few tens of thousands of years, possibly much longer.

Conclusions

Paleomagnetism can only record magnetic events and the interpretation of these magnetic events in terms of geologic events are in some instances highly subjective (just like radioactive age dating) and must rely on as many other sources of evidence as possible. For instance:

1. Paleopole positions of ore and host samples in the mines agree with each other. Their average pole (MV) is at 124° E and 45° N, with K = 176 and α_{95} = 5.8°.

2. Paleopole positions of hosts untouched by the mineralization are different from those in the mines. SQ has its pole at 132° E and 13° N whereas the VD pole is at 144° E and 38° N.

3. The apparent age of the magnetization of the rocks in the mine, estimated from the apparent polar wandering path for the North American stable craton, is around Early Permian-Late Pennsylvanian time.

4. The above paleomagnetic results may best be explained by the proposition that the paleomagnetic clock of the host rocks in the mine was reset by mineralization which would also be Early Permian-Late Pennsylvanian in age.

If the suggestion that long distance migration of fluids across a major part of southern Missouri is accepted as a working hypothesis, a considerable territory involved in possible migration routes could have commercial oil accumulations and, in dolostone settings, other ore fields that have yet to be discovered (Beales and Hardy, 1980; Friedman, 1980).

[2] We reject statements to the contrary by, e.g., Sverjensky, et al., 1979; Sverjensky, 1980; and McLimans, et al., 1980, but do not wish to digress here to debate the points.

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